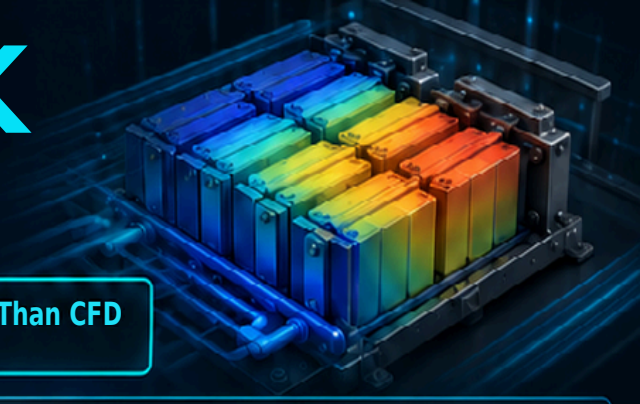
THERMAL PACK

Configurable Reduced-Order Battery Thermal Design & Optimization Platform



Independent Engineering Project

Up to 95% Faster Than CFD
for Iterative Studies

PROJECT SUMMARY

A configurable reduced-order electro-thermal platform for rapid battery thermal analysis, design and optimization. ThermalPack enables fast, accurate evaluation of thermal performance across a wide range of battery designs, cooling configurations, and operating conditions, bridging the gap between detailed CFD and early-stage design decisions.

- **Fast Parametric Studies**
- **Design Optimization**
- **Validated and Reliable**

KEY FEATURES & CAPABILITIES

Multiple Cell Formats
• Cylindrical • Prismatic • Pouch

Configurable Battery Architecture
Series-parallel layout, rows, topology, stacking direction, geometry scaling.

Inputs
Cell geometry, material, thermal properties, pack layout, cooling parameters.

Cooling Formats
Liquid (top / bottom, side, serpentine) and forced-air cooling architectures.

Electro-Thermal Models
Ohmic heat generation, 1RC, 2RC, transient load profiles.

Thermal KPIs
Peak and average transient cell temperatures, hotspots, safety margins, cell-to-cell ΔT, within-cell ΔT.

3D Visualization
Height-resolved or cell-level thermal visualization.

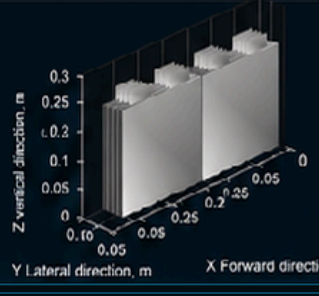
Results & Optimization
Dashboard plots, operating and design optimization, feasibility maps, exportable results.

WORKFLOW & KEY CAPABILITIES

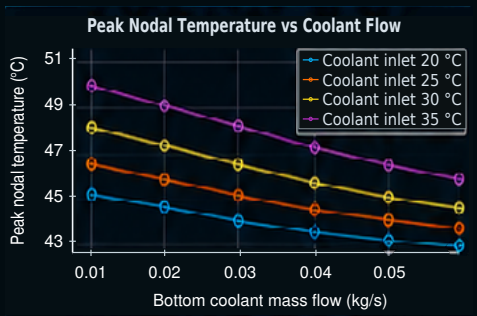
1 Configuration
Define battery design, thermal models and operating conditions.

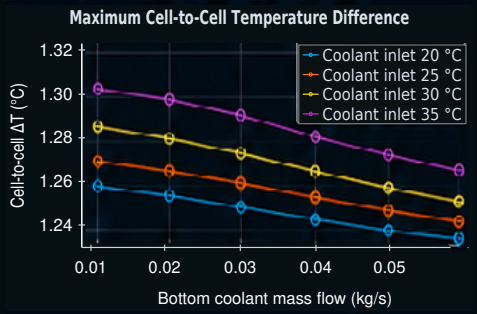
Battery & Simulation Configuration

Coolant flow (kg/s)	0.02
Top flow share (%)	50
Top flow (kg/s)	0
Bottom flow (kg/s)	0.02
Coolant inlet (°C)	25
Ambient (°C)	25
Cell → plate Rth (K/W)	1.2
Cell → ambient Rth (K/W)	25
Inter-cell heat transfer	Enabled
Inter-cell Rth (K/W)	1
Inter-parallel assembly Rth (K/W)	1

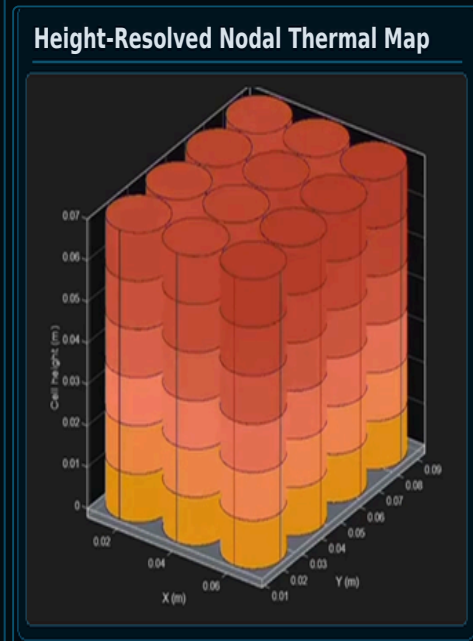


2 Thermal KPIs
Visualize performance, identify optimal operating points and feasible designs.





3 3D Visualization
Explore temperature distribution across the battery pack.



THERMALPACK

Optimization & Model Validation Highlights



Operating Optimization | Design Optimization | 3 Validation Cases

THERMAL OPTIMIZATION

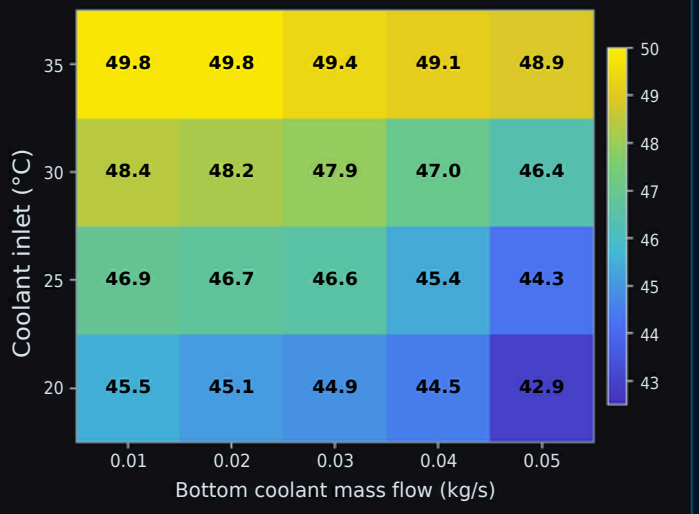
OPERATING OPTIMIZATION

Rapid assessment of the thermal operating envelope across different cooling conditions.

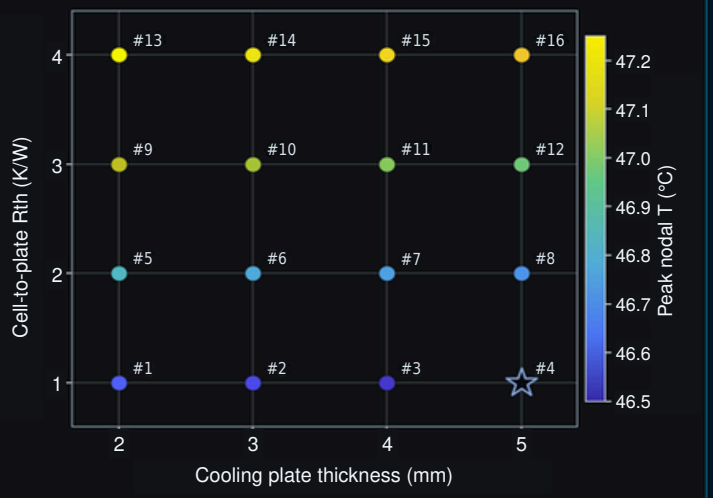
DESIGN OPTIMIZATION

Systematic screening of design configurations against thermal performance requirements.

Peak Nodal Temperature (°C)



Design Space — Peak Nodal Temperature



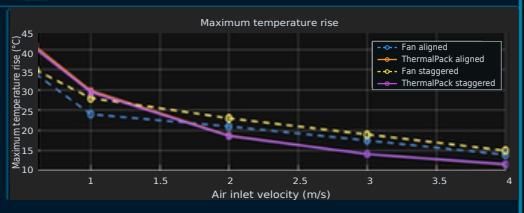
PROCESS

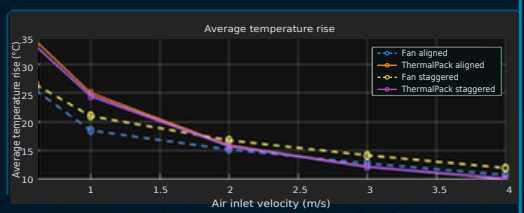
Parameter Sweeps → **KPI Comparison** → **Design Feasibility**

MODEL VALIDATION

Validated across forced-air and bottom-cooled benchmark cases.

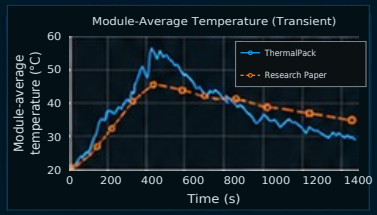
1 Fan 2019 — Forced-Air Cooling

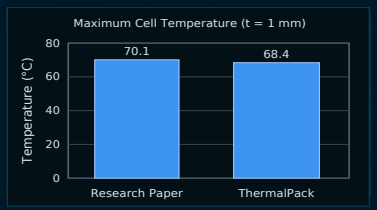




Maximum and average temperature-rise trends benchmarked against Fan et al. (2019) for aligned and staggered cylindrical arrays. ThermalPack captures airflow-dependent cooling trends, with RMSE: 3.6–4.8 °C.
2C | 20 °C inlet | Air velocity: 0.6–4 m/s

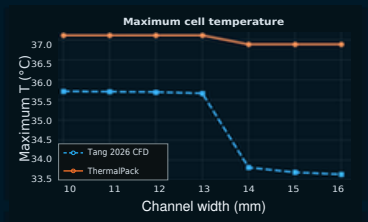
2 Bahiraei 2017 — Bottom Cooling

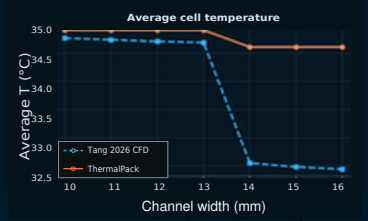




Prismatic NCA module benchmarked against Bahiraei et al. (2017) under indirect bottom cooling. ThermalPack captures transient heating and cooling trends, with a peak cell temperature of 68.4 °C versus 70.1 °C in the research paper (−2.4%).
6P Module | 1400 s FUDC | 1 mm plate

3 Tang et al. 2026 — Cooling-Plate Benchmarking



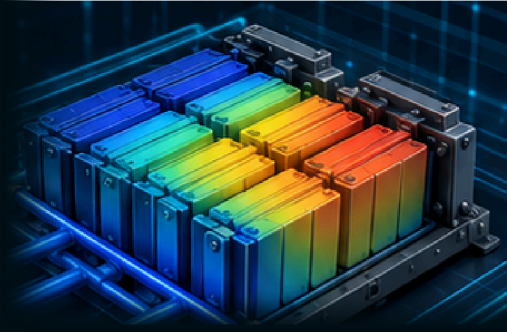


Reduced-order cooling-plate model benchmarked against Tang et al. (2026). ThermalPack captures the general influence of channel width on maximum and average cell temperatures.
5S LFP | 2C | 1800 s | 10–16 mm

From thermal insight to smarter battery design. | Fast | Configurable | Optimization-Ready | Validation-Driven

THERMALPACK

References for Model Validation



VALIDATION REFERENCES

1 Fan 2019 - Forced-Air Cooling

Yuqian Fan, Yun Bao, Chen Ling, Yanyan Chu, Xiaojun Tan, Shuting Yang.

Experimental study on the thermal management performance of air cooling for high energy density cylindrical lithium-ion batteries

Applied Thermal Engineering, 155 (2019), pp. 96-109.

DOI: 10.1016/j.applthermaleng.2019.03.157

2 Bahiraei 2017 - Bottom Cooling

Farid Bahiraei, Amir Fartaj, Gholam-Abbas Nazri.

Electrochemical-thermal Modeling to Evaluate Active Thermal Management of a Lithium-ion Battery Module

Electrochimica Acta, 254 (2017), pp. 59-71.

DOI: 10.1016/j.electacta.2017.09.084

3 Tang et al. 2026 - Cooling-Plate Benchmarking

Youming Tang, Jialong Qin, Jiawei Chen, Xubin Song, Guangbin Deng, Leyu Wang.

Thermo-Hydraulic Performance and Energy-Efficiency Analysis of a Multiparameter Serpentine Cold Plate for Prismatic LFP Battery Modules

Modelling and Simulation in Engineering, 2026(1), article 4032488, 14 pp.

DOI: 10.1155/mse/4032488